Supplementary Material

Position-Faithful Merge Tree Maps: Certified Bounds and Near-Optimal Layouts

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Theorem numbers refer to the paper. Notation: T_t is the merge tree of step t (join convention; apply everything to $-f$ for superlevel sets), $V_{ij} = f(\mathrm{lca}(i,j))$, $q_i \in [0,L]$ the reference of leaf i, W_i its interval width, g the gap, $H_i = \rho W_i/2$ the anchor slack, d_{top} the labelled merge error (Eq. 2 of the paper) and Δ_F the scalar range. All claims below were also checked against exhaustive enumeration on random trees (`code/tests/test_order_certificate.py`, `code/tests/test_interval_order.py`).

S1 Proofs

Interval model. For a leaf order π, leaf k occupies $[c_k - W_k/2, c_k + W_k/2] \subseteq [0,L]$; consecutive intervals do not overlap and are at least g apart; the anchor satisfies $|x_k - c_k| \le H_k$. If leaf a precedes leaf b and every leaf of a set $S \ni a,b$ lies between them (inclusive), then

$$c_b - c_a \;\ge\; \sum_{k\in S} s_k - g - \tfrac{W_a}{2} - \tfrac{W_b}{2}, \qquad s_k = W_k + g, \tag{1}$$

because the half intervals of a and b, the full intervals of the leaves between them and one gap per consecutive pair fit between the two centres; leaves of the order that are not in S only add distance. The attained cost is $\tau(\pi) = \min \max_k |x_k - q_k|$ over all such placements.

Theorem 1 (Theorem 2 of the paper)**.** *Let $\ell_k = q_k - W_k/2 - H_k$, $h_k = q_k + W_k/2 + H_k$, and for a leaf set S let $A(S) = \max_\alpha[\alpha + \sum_{k\in S,\ell_k\ge\alpha} s_k]$ and $B(S) = \max_\beta[\sum_{k\in S,h_k\le\beta} s_k - \beta]$, the maxima taken over $\alpha \in \{\ell_k\}$, $\beta \in \{h_k\}$. (i) If every leaf of X precedes every leaf of Y, then $\tau \ge (A(X) + B(Y) - g)/2$. (ii) Every placement satisfies $\tau \ge B(\mathrm{all}) - g$ and $\tau \ge A(\mathrm{all}) - g - L$. (iii) With $W \equiv 0$ and $g = 0$, the bound of (i) over sibling pairs equals $L\nu^*$.*

Proof. (i) Take $S_X \subseteq X$, $S_Y \subseteq Y$, let a be the first leaf of S_X and b the last leaf of S_Y in the order. All of $S = S_X \cup S_Y$ lies between them, so by (1) and the anchor slack, $x_b - x_a \ge \sum_S s_k - g - W_a/2 - W_b/2 - H_a - H_b$. Since $\tau \ge q_a - x_a$ and $\tau \ge x_b - q_b$,

$$\tau \;\ge\; \tfrac{1}{2}\big[(q_a - x_a) + (x_b - q_b)\big] \;\ge\; \tfrac{1}{2}\Big[\ell_a - h_b + \sum_{k\in S} s_k - g\Big] \;\ge\; \tfrac{1}{2}\Big[\min_{S_X}\ell + \sum_{S_X} s_k + \sum_{S_Y} s_k - \max_{S_Y} h - g\Big].$$

For $S_X = \{k \in X : \ell_k \ge \alpha\}$ the first two terms are at least $\alpha + \sum_{k\in X,\ell_k\ge\alpha} s_k$; for $S_Y = \{k \in Y : h_k \le \beta\}$ the last ones are at least $\sum_{k\in Y,h_k\le\beta} s_k - \beta$. The two choices are independent, so maximizing each gives $A(X)$ and $B(Y)$. Each maximum is over upper (resp. lower) sets of one sorted list and is computed by one sort and a prefix sum. (ii) Let b be the last leaf of a set S. The intervals of S lie in $[0, c_b + W_b/2]$, so $c_b \ge \sum_S s_k - g - W_b/2$, $x_b \ge c_b - H_b$ and $\tau \ge x_b - q_b \ge \sum_S s_k - g - h_b \ge \sum_S s_k - g - \max_S h$. Lower sets of h give $B(\mathrm{all}) - g$. The right end is symmetric with the first leaf of S and gives $A(\mathrm{all}) - g - L$. (iii) Then $\ell = h = q$, $s = 0$, $A(X) = M_X$, $B(Y) = -m_Y$, and $\min(A(X) + B(Y), A(Y) + B(X))/2 = \min(M_X - m_Y, M_Y - m_X)/2$, which is L times the term of ν^*. □

Lemma 1 (Separated sets)**.** *If no element of Q lies strictly between two elements of P in a linear order, and no element of P lies strictly between two elements of Q, then all of P precedes all of Q or vice versa.*

Proof. All of Q lies outside the span of P. If Q had elements on both sides of that span, all of P would lie strictly between two elements of Q. □

Theorem 2 (Theorem 3 of the paper)**.** *Let $A_\varepsilon(u)$ be the highest ancestor of u with $f(A_\varepsilon(u)) \le f(u) + 2\varepsilon$. In every map with $d_{\mathrm{top}} \le \varepsilon$: (i) no leaf outside $A_\varepsilon(u)$ lies between two leaves of u; (ii) if no leaf of w is in $A_\varepsilon(u)$ and no leaf of u is in $A_\varepsilon(w)$, then all leaves of u precede all leaves of w or vice versa. Consequently* $\mathrm{NROE} \ge \frac{1}{2L}\max\min(M_u - q_k, q_k - m_u)$ *over leaves $k \notin A_\varepsilon(u)$,* $\mathrm{NROE} \ge \frac{1}{2L}\max\min(M_u - m_w, M_w - m_u)$ *over pairs as in (ii), and the same pairs give width-aware bounds on τ through Theorem 2(i), with $A(\{k\}) = \ell_k + s_k$ and $B(\{k\}) = s_k - h_k$ for a single leaf.*

Proof. (i) For a leaf k outside $A_\varepsilon(u)$ and any leaf i of u, $\mathrm{lca}(i,k)$ is a proper ancestor of $A_\varepsilon(u)$, so $V_{ik} > f(u)+2\varepsilon$ by the choice of $A_\varepsilon(u)$; by Lemma 1 of the paper, k cannot lie between two leaves of u. (ii) By (i) applied to u and to w, the conditions of Lemma 1 hold for their leaf sets. The bounds follow as in the proofs of Theorems 1 and 2(i), since those only used that one set precedes the other. □

The coarse bound. If a node u has parent level above $f(u)+2\varepsilon$, then $A_\varepsilon(u) = u$ and u is contiguous by Theorem 3(i). Hence in every map with $d_{\mathrm{top}} \le \varepsilon$ the children of each node of $T^{(2\varepsilon)}$ (all nodes within 2ε of their parent flattened) occupy disjoint blocks, and Theorems 1 and 2 apply to $T^{(2\varepsilon)}$. The chain bound contains every pair used here and is never weaker.

Data resolution. Treating levels closer than $\varepsilon_0\Delta_F$ as equal is the statement $d_{\mathrm{top}} \le \varepsilon_0\Delta_F$ for the ties; all bounds hold with $\varepsilon_0+\varepsilon$ in place of ε.

Proposition 1 (Optimal barriers)**.** *Fix an order π and the extremum values. For the gap k between the kth and $(k{+}1)$st leaf let $C_k = \min\{V_{\pi(i)\pi(j)} : i \le k < j\}$ and $F_k = \max(f(\pi(k)), f(\pi(k{+}1)))$. The smallest achievable merge error is*

$$\delta^*(\pi) = \max\Big(0,\ \max_{i<j} \tfrac{1}{2}\big(V_{\pi(i)\pi(j)} - \max_{i\le k<j} C_k\big),\ \max_k (F_k - C_k)\Big),$$

attained by the barriers $b_k = C_k + \delta^*(\pi)$.

Proof. The displayed merge of the ith and jth leaf is $\max_{i\le k<j} b_k$, and each barrier must be at least F_k to keep both basins. An error δ requires $b_k \le V_{\pi(i)\pi(j)} + \delta$ for every pair spanning k, i.e., $b_k \le C_k + \delta$; it also requires $\max_{i\le k<j} b_k \ge V_{\pi(i)\pi(j)} - \delta$ and $b_k \ge F_k$. The largest admissible barriers $b_k = C_k + \delta$ satisfy all constraints exactly when δ is at least each term of δ^*; smaller barriers only lower the maxima. Hence δ^* is optimal and attained. □

S2 Methods and protocol

Ordering details. Candidate trees: the exact tree; one tree per distinct parent gap $s \le 2(\varepsilon_0+\varepsilon)$ with every node within s of its parent flattened; and a conflict-driven flattening that repeatedly releases one or both nodes of the sibling pair with the largest order cost if the chain condition of Theorem 3 holds and the exact δ^* of the resulting midpoint order stays within the budget. The four best candidates by initial τ, plus the exact tree, are refined by local search: for every node, all insertions of one child block at another position (all of them up to eight children, otherwise within distance three), accepted when they lower $(\tau, \sum \text{positive terms}^2)$ and keep $\delta^* \le (\varepsilon_0+\varepsilon)\Delta_F$; at most eight passes. On 300 random trees with widths, this search reached the exhaustive optimum of τ on 296; the point-model order on 250.

Metrics. SNS, TW ($k=1$), TD and ME follow the protocol of the completed nine-input table of our earlier evaluation and of ST-MTM: unweighted support centroids in the native spatial dimension; positions on the full $[0,1]$ canvas; motion events over two steps whose source distance changes by more than 2% of the domain diagonal, judged with a display threshold of 2% of the occupied range (fixed canvas for Cylinder). Planar domain diagonals are not stored; we use the midpoint of the interval that reproduces the stored event count, and thresholds are nested, so the whole interval selects the same events. Recomputing the stored baselines from their saved anchors agrees exactly in SNS and TD except Cylinder (relative 10^{-5}), in TW to the third or fourth decimal on Tangaroa and Cylinder, and in ME up to one event on $PM_{2.5}$, Ring and Cylinder; the stored values were taken from rendered anchors.

Instances. Given the native branch decisions, the TMTM linearization matched the pinned official C++ core in all 3127 arc intervals and anchors of the five planar sequences (687, 431, 289, 463, 1257), with zero merge error. Native ST-MTM anchors and intervals satisfied the KKT conditions of ST-MTM Eqs. 1–2 with relative residuals at most $3.6\cdot10^{-8}$ and $9.6\cdot10^{-7}$ (planar) and stationarity residuals at most $7.1\cdot10^{-11}$ (volumetric). SciPy optimal leaf ordering exceeded the exhaustive optimum of its adjacency objective on 1, 7, 0, 2 and 0 frames of Storms, Wildfires, OLR, $PM_{2.5}$ and Ring. Enumerating all 2^k branch decisions against the same previous greedy layout, TMTM's greedy step was optimal at 36/39, 47/59 and 351/365 steps of Ring, OLR and $PM_{2.5}$; summed over steps it exceeded the per-step optimum by 17.4%, 5.7% and 2.2%.

Labelled interleaving. We checked the conditions of the identification $d_{\mathrm{top}} = d_I^L$ against arXiv:1908.00063: labels must cover all leaves (labels on inner vertices are allowed), the induced matrix includes the diagonal $f(\pi(i))$, and d_I^L is defined as the ℓ_∞ distance of induced matrices. Every column of the rendered Ring and OLR maps has exactly one local extremum per feature (0 exceptions in 200 columns), so no unlabelled leaf occurs there.

Runtime. Apple M2 Pro (12 cores, 32 GB), macOS 26.6, Python 3.12.7, NumPy 2.0.2, SciPy 1.13.1, one process. Per frame: bounds 0.11–0.62 ms (planar) and 2.7–5.5 ms (volumetric); ordering 0.40–3.7 ms and 11–103 ms; capped QP 2.3–9.0 ms and 0.25–0.50 s.

S3 Controlled hierarchy-conflict experiment (N4)

This experiment isolates reference-axis conflict; its sweep is not a physical time sequence. The input is a labelled four-leaf join tree, with leaf values zero, sibling pairs $\{A,B\}$ and $\{C,D\}$ at level a, and root level one. Its cophenetic matrix has diagonal zero, within-pair entries a, and cross-pair entries one. The canvas and scalar range are both one. References are $q = (0.2, 0.5, 0.5-d, 0.8-d)$, so $d = \max q_X - \min q_Y$ is a signed overlap: negative values separate the hulls. Since $d \le 0.3$, the alternative orientation costs at least as much, giving $\nu^* = \max(0,d)/2$.

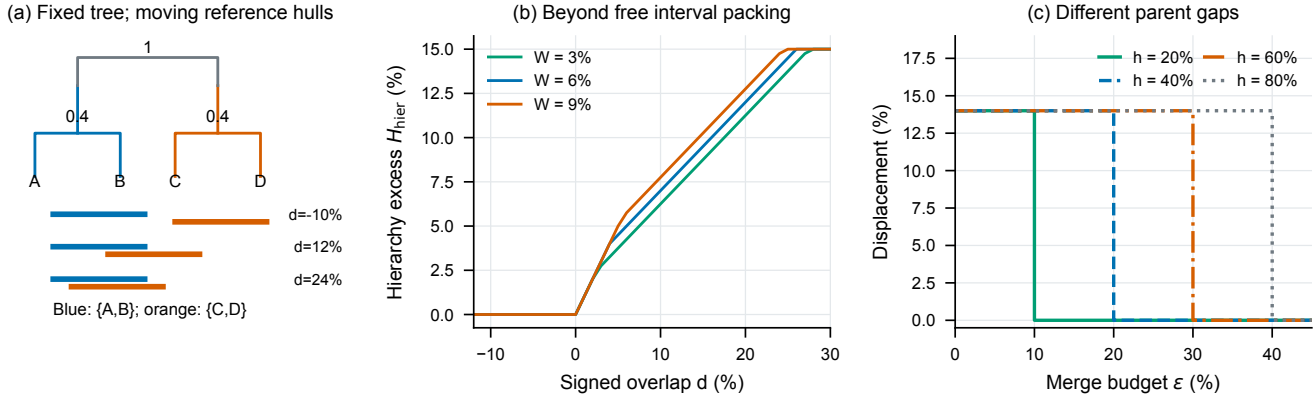


Figure S1: N4 protocol and additional results. (a) The fixed tree and three examples of moving reference hulls. (b) Extra displacement caused by hierarchy, after subtracting the optimum with free interval packing, for three positive widths. (c) Budget thresholds for four parent gaps: release occurs at half the gap.

Overlap and width sweep. Fix $a = 0.4$. Use 43 equally spaced values $d = -0.12, -0.11, \ldots, 0.30$, and equal widths $W \in \{0, 0.03, 0.06, 0.09\}$ (172 settings). Set $\rho = 0.5$; use zero gap in the point model and gap $g = 0.01$ for positive widths. No tolerance budget, temporal term or quadratic secondary objective is used: $\varepsilon_0 = \beta = 0$. For each setting enumerate all $4! = 24$ orders, compute their exact optimal barrier errors, and independently solve the continuous minimax LP in variables (x_i, c_i, t) with objective $\min t$, interval nonoverlap, canvas boundaries, $|x_i - c_i| \le \rho W/2$, and $|x_i - q_i| \le t$. The LPs include the zero-width non-strict point model. Select the minimum LP value among zero-merge-error orders, denoted τ_{exact}, and among all orders, denoted τ_{free}.

The point bound and width bound are attained throughout this family. Specifically,

$$\tau_{\text{exact}} = \max\big(0, [d + (1-\rho)W + g]/2\big).$$

The largest discrepancy between independently solved LP costs and the fixed-order closed form over all 4128 overlap-sweep LPs is $1.67 \cdot 10^{-16}$ of the canvas. The width-bound gap and width-search gap are below $1.1 \cdot 10^{-16}$. This validates these controlled cases; it does not establish tightness for arbitrary trees or width assignments.

Hierarchy cost versus packing. Figure S1b reports $H_{\text{hier}} = \tau_{\text{exact}} - \tau_{\text{free}}$, keeping widths, gaps, slack and canvas identical. This diagnostic separates the extra cost of exact hierarchy from free interval packing; it is distinct from the anchor slack H_i. At $d = 0.24$, $W = 0.06$, the free optimum is zero and the hierarchy excess is 0.14. At $d = 0.30$, two pairs of references coincide; free packing itself costs 0.02, compared with the exact cost 0.17, so the hierarchy excess is 0.15.

Budget and parent-gap sweep. Fix $d = 0.24$, $W = 0.06$, $g = 0.01$, $\rho = 0.5$. Vary $a \in \{0.2, 0.4, 0.6, 0.8\}$, giving parent gaps $h = 1 - a \in \{0.8, 0.6, 0.4, 0.2\}$. At each of 91 budgets $\varepsilon = 0, 0.005, \ldots, 0.45$, select the minimum LP cost among orders with $\delta^*(\pi) \le \varepsilon$ (364 settings). The 24 spatial LPs are shared across these settings, for 4152 independent LP solves in total. The width-aware search reaches the exhaustive optimum at every budget. Interleaving first becomes feasible at $\varepsilon = h/2$, as predicted by the 2ε separation condition. Below that threshold the best displacement is 0.14; at and above it the free order

A,C,B,D attains zero displacement, with optimal barriers $(a+h/2, a+h/2, a+h/2)$ and measured merge error $h/2$. All selected barriers are checked directly against all pairwise merge levels.

Reproduction. The new LP checks use Python 3.10.19, NumPy 2.0.2 and SciPy 1.13.1. Run `code/experiments/controlled_conflict/run.py` for the exhaustive/LP checks and saved CSV/JSON records; `plot.py` in the same directory reads those records to generate both figures. The experiment uses the paper's existing certificate, barrier-cost and width-search implementations; the minimax LP is assembled independently of the fixed-order cost formula.

S4 Local response to a larger budget

Question and definition. Reducing an aggregate position error need not improve every feature. For a fixed feature i at the same time t, let $e_{i,t}(\varepsilon) = |x_{i,t}(\varepsilon) - q_{i,t}|/L$ and $\Delta e_{i,t} = e_{i,t}(\varepsilon_b) - e_{i,t}(\varepsilon_a)$, where $\varepsilon_a < \varepsilon_b$. Positive changes mean worsening. These compare budgets, not motion between time steps. Define $R_h = \#\{(i,t) : \Delta e_{i,t} > h\}/N$. For the same feature–frame population,

$$E_{\text{pool}}(\varepsilon_b) - E_{\text{pool}}(\varepsilon_a) = N^{-1} \sum_{i,t} \Delta e_{i,t}.$$

Positive and negative local changes can therefore coexist while their mean decreases.

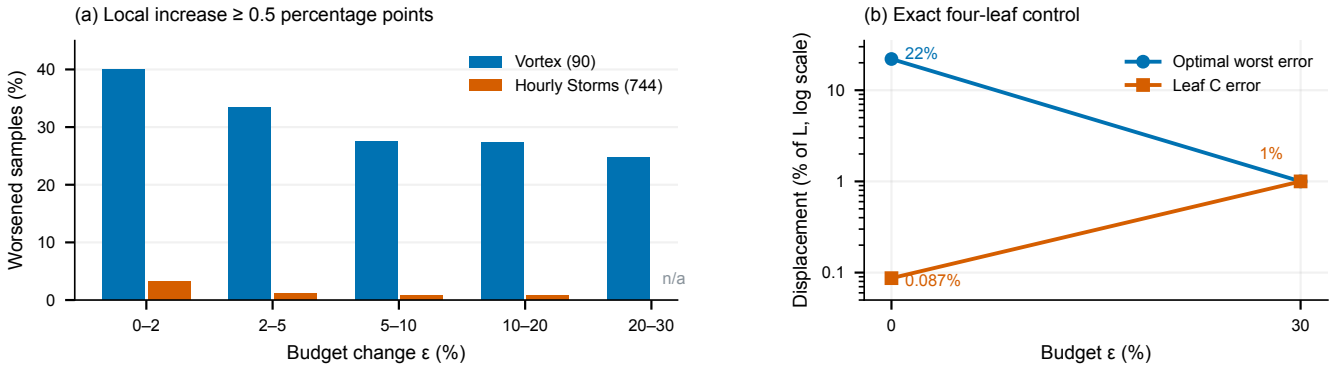


Figure S2: Local response to larger merge budgets. (a) Percentage of fixed feature–frame observations whose position error increases by at least 0.5 percentage points of the canvas. Vortex has 90 frames; the separate hourly Storms diagnostic has 744. Hourly Storms has no 30% result (n/a). These cohorts do not replace the seven-sequence benchmark and do not isolate dimensionality. (b) An exhaustive four-leaf control: the optimal maximum displacement falls from 22% to 1%, while leaf C's error rises from 0.087% to 1%. The vertical scale is logarithmic.

Nested feasibility and pointwise error. With input, reference, interval geometry and any externally imposed position cap fixed, increasing the merge budget enlarges the feasible set. The globally minimal worst displacement cannot increase, but may plateau. This is a statement about an optimal value, not a componentwise inequality between the errors of two selected layouts. The implemented order search is also heuristic and its secondary objective balances geometry and temporal behavior. The following control separates the pointwise issue from both heuristic search and temporal history.

Exhaustive control. Use a static four-leaf join tree: leaf values zero, sibling pairs $\{A,B\}$ and $\{C,D\}$ merging at 0.4, root at 1. Set $L = \Delta_F = 1$, references $q = (0.20, 0.80, 0.18, 0.60)$, widths $W_i = 0.06$, gap $g = 0.01$, anchor slack $\rho = 0.5$ and $\beta = 0$. Source centres are (q_i, y_i) with $y = (0.144, 0.156, 0.146, 0.052)$. Enumerate all 24 orders and independently solve the minimax placement LP for each, using the canvas, interval, separation and anchor-slack constraints. Among minimax-optimal orders, minimize the same fixed convex secondary objective with reference, geometry and eccentricity weights 4, 1 and 0.1, and no temporal term. A $10^{-9}L$ allowance is used only for numerical cap feasibility; independent LP first-order oracles certify convex QP gaps below 10^{-8}.

At $\varepsilon = 0$, eight exact orders are eligible. The unique minimax-optimal order is C,D,A,B with $\tau^* = 0.22$. The reference-hull/width bound is $(0.60 - 0.20 + 0.04)/2 = 0.22$. At $\varepsilon = 0.30$, all 24 orders are eligible. The unique minimax-optimal order is C,A,D,B with $\tau^* = 0.01$. The references of C and A are separated by 0.02, but their anchors require separation $(1-\rho)W + g = 0.04$, certifying the lower bound $(0.04 - 0.02)/2 = 0.01$. Feasible placements attain both bounds. Original and relaxed barriers are $(0.4, 1, 0.4)$ and $(0.7, 0.7, 0.7)$; all six labelled pairwise merge errors are remeasured. C's selected

errors are 0.000868919 and 0.010000001, i.e., 0.087% and 1% of L. Thus strict improvement of the exact global minimax value need not improve every point. This constructed counterexample establishes lack of guarantee, not prevalence.

Additional diagnostic cohorts. Vortex uses 90 original 128^3 normalized scalar fields, 111–156 retained extrema per frame and 12,446 feature–frame observations. Its supplied scalar range is $[-1,1]$; the join convention uses $-f/2$. A fixed 2% persistence simplification precedes layout and differs from the budget. References are fixed X voxel coordinates on $L = 127$. The hourly Storms diagnostic uses 744 hourly ERA5 anomaly fields for December 1999 on the retained 49×49 analysis grid, with 1–12 retained minima per frame (3,502 observations). The anomaly subtracts a centred 192-hour per-location background; spatial processing and reference are unchanged. This differs from the main paper's 118 half-day Storms sequence. These separately prepared cohorts are not added to the 917-frame cross-method population. Physical feature identity beyond the saved computational matching is not assumed.

Both use the frozen certified order/optimal-barrier/capped-QP construction, fixed widths, gaps, references and feature sets, with $B_t = \tau_t + \beta$. Vortex budgets are $\{0, 0.02, 0.05, 0.10, 0.20, 0.30\}$; hourly Storms uses $\{0, 0.02, 0.05, 0.10, 0.20\}$. Comparisons pair the same feature at the same time, never by proximity after layout. The additional 30% Vortex run covers all 90 frames from the initial frame, preserving temporal history, and passed independent source-tree, interval, cap, profile and sampled-LP checks. This audit remeasures saved continuous anchors; no layout or solver weight was tuned for its results.

Reporting thresholds. Changes beyond $10^{-6}L$ are classified as positive or negative; the rest are unchanged. Increases of at least 0.005 and 0.01 of L (0.5 and 1 percentage points) are also reported. These are numerical reporting thresholds, not calibrated perceptual limits. Counts are descriptive: repeated features and times are not treated as independent trials, and no inferential confidence intervals are claimed. Table S1 reports all adjacent budget changes.

Table S1: Fixed-population budget comparisons. N is feature–frame observations per transition. "Worse" uses $\Delta e > 10^{-6}$; pp means absolute percentage points of L.

Cohort	$\varepsilon_a \to \varepsilon_b$	N	Worse (%)	≥ 0.5 pp (%)	≥ 1 pp (%)
Vortex	0%–2%	12,446	43.27	39.96	36.46
Vortex	2%–5%	12,446	40.29	33.42	29.29
Vortex	5%–10%	12,446	37.47	27.47	23.06
Vortex	10%–20%	12,446	32.77	27.33	24.02
Vortex	20%–30%	12,446	30.91	24.72	20.60
Hourly Storms	0%–2%	3,502	7.14	3.20	2.11
Hourly Storms	2%–5%	3,502	4.45	1.17	0.66
Hourly Storms	5%–10%	3,502	2.51	0.86	0.57
Hourly Storms	10%–20%	3,502	3.71	0.86	0.51

At 20% to 30% on Vortex, pooled error falls from 8.794% to 4.273%, and the sequence maximum from 30.769% to 28.741%, but 3,847 of 12,446 observations worsen (30.91%); 3,077 worsen by at least 0.5 percentage points (24.72%). Restricting to the 88 frames whose mean and maximum both strictly decrease leaves 12,141 observations, of which 3,778 worsen (31.12%). The effect is therefore not explained solely by frames whose overall position error worsens. Local increases also occur with an unchanged order at that time (13 observations in this transition); different temporal histories can affect a fixed-order QP, so this is not a causal decomposition of order and temporal effects.

Illustrative case and interpretation. At frame 42 (zero-based), track 138's error grows from 0.031% to 24.800% of L from 20% to 30%, although the frame mean falls from 12.332% to 5.579% and the maximum from 28.467% to 25.719%. The case-selection rule is explicit in Fig. S3; full-population counts, not this extreme, support prevalence statements. An analyst choosing a budget for a feature of interest should inspect its signed local response alongside aggregate displacement and hierarchy-error disclosure. This diagnostic changes no certified constraint and claims no pointwise improvement guarantee.

Precision example retained from the main text. The data resolution should match field precision. On Cylinder, whose saddles are nearly degenerate, ν^* is 12.24% when only exactly equal levels are merged and 6.15% at $\varepsilon_0 = 10^{-6}$; part of this hierarchy lies below data precision. This precision control differs from increasing the analyst budget with a fixed feature population.

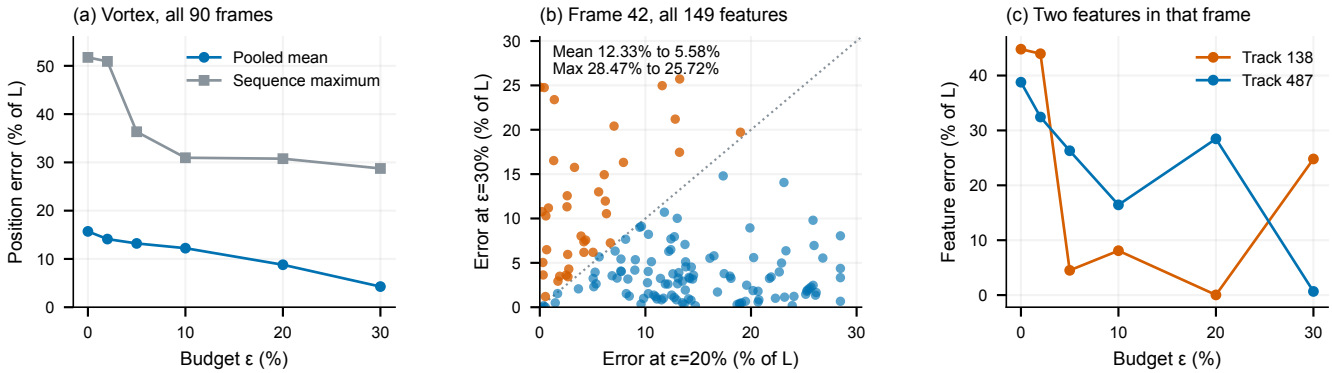


Figure S3: Vortex response to changing budget. (a) Pooled and maximum errors over all 90 frames. (b) All 149 features at frame 42, comparing 20% and 30%. Orange indicates an increase of at least 0.5 percentage points; points above the diagonal worsen. Both frame mean and maximum decrease. (c) Two features from that frame: track 138 worsens and track 487 improves from 20% to 30%. The frame is selected by the largest local increase among frames whose mean and maximum both decrease; it illustrates an extreme, not typical behavior.

Reproduction. Run `code/experiments/local_redistribution/run.py`, then `control.py` and `plot.py` in that directory. Bound compressed inputs, SHA256 manifests, all 76,238 pointwise transition records, summaries and the 24-order control are in `code/results/local_redistribution20261030/`. Inputs contain continuous anchors, references, stable track IDs and orders; this audit needs no external volume archive and does not rerun the original volume extraction.

S5 Additional results

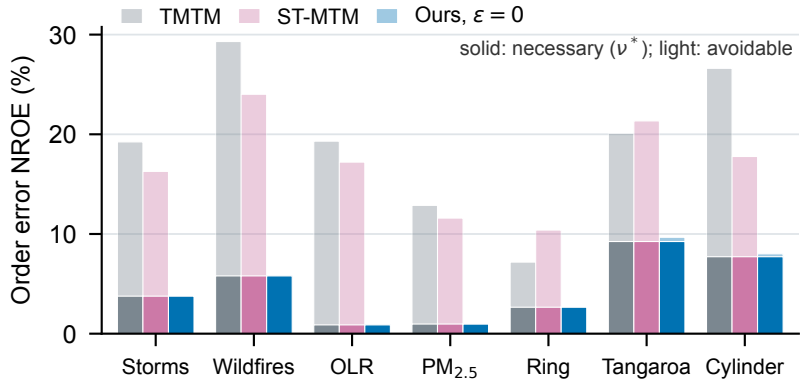


Figure S4: Mean order error per sequence split into the necessary part ν^* (solid) and the avoidable remainder (light).

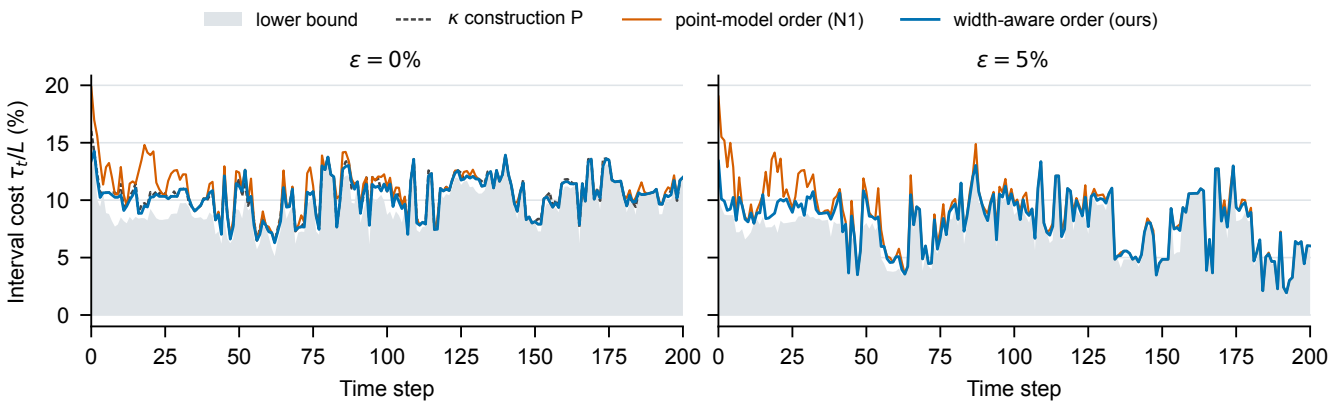


Figure S5: Tangaroa, 201 steps: interval cost τ_t/L of the point-model order (orange), the width-aware order (blue) and the κ construction (dashed), with the certified lower bound (grey).

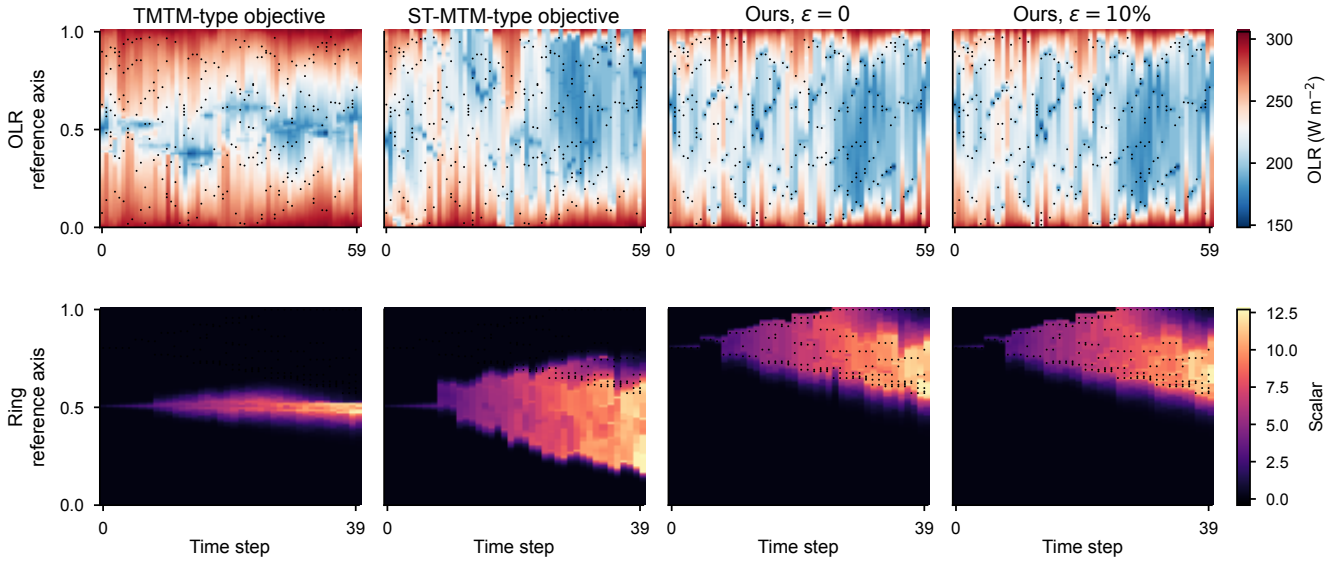


Figure S6: OLR (60 days) and Ring (40 steps) maps that differ only in the layout objective; dots mark references.

Table S2: All values of the cross-method comparison (paper Fig. 6), in percent except SNS and TW.

Sequence	Method	NROE	worst merge	ME	SNS	TW
Storms	TMTM	19.23	0.00	47.1	0.080	0.824
	ST-MTM	16.28	0.00	41.3	0.030	0.965
	κ R20 (earlier)	1.16	26.07	18.3	0.110	0.755
	Projection	0.00	58.66	14.4	0.106	0.723
	Ours, $\varepsilon = 0$	3.77	0.00	24.0	0.120	0.770
	Ours, $\varepsilon = 10\%$	1.43	8.51	19.2	0.113	0.752
Wildfires	TMTM	29.30	0.00	39.3	0.221	0.817
	ST-MTM	24.02	0.00	35.7	0.079	0.919
	κ R20 (earlier)	0.34	19.56	31.2	0.058	0.872
	Projection	0.00	52.58	33.0	0.057	0.861
	Ours, $\varepsilon = 0$	5.80	0.00	34.8	0.085	0.875
	Ours, $\varepsilon = 10\%$	1.09	8.35	34.8	0.061	0.876
OLR	TMTM	19.31	0.00	44.6	0.142	0.814
	ST-MTM	17.21	0.00	21.4	0.006	0.975
	κ R20 (earlier)	0.00	33.56	5.4	0.009	0.963
	Projection	0.00	33.56	5.4	0.009	0.961
	Ours, $\varepsilon = 0$	0.89	0.00	5.4	0.012	0.956
	Ours, $\varepsilon = 10\%$	0.30	5.15	7.1	0.010	0.962
$PM_{2.5}$	TMTM	12.87	0.00	45.3	0.051	0.760
	ST-MTM	11.60	0.00	51.6	0.026	0.922
	κ R20 (earlier)	0.01	4.67	49.5	0.052	0.729
	Projection	0.00	5.62	49.5	0.052	0.725
	Ours, $\varepsilon = 0$	0.98	0.00	44.2	0.065	0.751
	Ours, $\varepsilon = 10\%$	0.00	5.62	49.5	0.052	0.727
Ring	TMTM	7.18	0.00	64.5	0.171	0.707
	ST-MTM	10.39	0.00	34.7	0.080	0.917
	κ R20 (earlier)	0.13	12.35	12.9	0.142	0.715
	Projection	0.00	12.35	12.1	0.143	0.698
	Ours, $\varepsilon = 0$	2.66	0.00	31.5	0.164	0.686
	Ours, $\varepsilon = 10\%$	0.05	9.40	11.3	0.142	0.707
Tangaroa	TMTM	20.09	0.00	97.5	0.525	0.670
	ST-MTM	21.34	0.00	70.0	0.165	0.814
	κ R20 (earlier)	0.02	31.02	30.5	0.335	0.746
	Projection	0.00	31.02	30.5	0.335	0.748
	Ours, $\varepsilon = 0$	9.67	0.00	52.2	0.362	0.726
	Ours, $\varepsilon = 10\%$	4.76	9.97	41.3	0.341	0.754
Cylinder	TMTM	26.62	0.00	67.0	0.296	0.802
	ST-MTM	17.77	0.00	46.0	0.135	0.899
	κ R20 (earlier)	0.08	14.28	18.9	0.040	0.891
	Projection	0.00	14.28	19.3	0.043	0.878
	Ours, $\varepsilon = 0$	8.01	0.00	33.1	0.110	0.814
	Ours, $\varepsilon = 10\%$	0.01	9.71	19.5	0.041	0.889

Table S3: Order error and position (percent). ν^* and mean NROE of native TMTM, ST-MTM and our exact layout; mean and worst displacement of our capped layouts; worst displacement and worst merge error (NMLE max) at $\varepsilon = 2\%$ and 10%.

				NROE			$\varepsilon = 0$		$\varepsilon = 2\%$		$\varepsilon = 10\%$	
Sequence	Frames	Leaves	ν^*	TMTM	ST-MTM	Ours	E_{pool}	E_∞	E_∞	NMLE	E_∞	NMLE
Storms	118	1–8	3.77	19.23	16.28	**3.77**	4.19	26.19	22.85	1.50	18.46	8.51
Wildfires	30	6–11	5.80	29.30	24.02	**5.80**	3.24	20.48	8.71	1.66	7.20	8.35
OLR	60	2–8	0.89	19.31	17.21	**0.89**	1.00	17.14	17.14	1.79	17.14	5.15
$PM_{2.5}$	366	1–7	0.97	12.87	11.60	**0.98**	1.87	20.26	19.75	1.81	5.86	5.62
Ring	40	1–7	2.66	7.18	10.39	**2.66**	2.44	11.90	7.56	1.65	2.79	9.40
Tangaroa	201	18–93	9.24	20.09	21.34	**9.67**	5.01	15.10	14.55	2.00	13.08	9.97
Cylinder	102	4–75	7.72	26.62	17.77	**8.01**	3.76	27.00	5.27	1.67	5.27	9.71

Table S4: Volumetric sequences (percent). κ P (exact) and R2 (relaxed, no merge-error cap): earlier construction. N1: certified order by NROE only. Jump: mean $|\Delta x - \Delta q|/L$ of matched features.

	Layout	E_{pool}	E_∞	NROE	jump	NMLE max
Tangaroa	κ P	5.01	16.86	9.61	2.59	0
Tangaroa	N1, $\varepsilon = 0$	4.92	20.69	9.24	2.73	0.00
Tangaroa	Ours, $\varepsilon = 0$	5.01	15.10	9.67	2.82	0.00
Tangaroa	κ R2	4.16	14.65	8.56	2.49	no cap
Tangaroa	N1, $\varepsilon = 2\%$	4.49	20.69	8.23	2.60	2.00
Tangaroa	Ours, $\varepsilon = 2\%$	4.55	14.55	8.60	2.76	2.00
Cylinder	κ P	5.40	32.08	12.26	5.87	0
Cylinder	N1, $\varepsilon = 0$	3.76	27.00	8.01	3.83	0.00
Cylinder	Ours, $\varepsilon = 0$	3.76	27.00	8.01	3.89	0.00
Cylinder	κ R2	0.62	5.27	0.25	0.47	no cap
Cylinder	N1, $\varepsilon = 2\%$	0.64	5.73	0.17	0.46	1.84
Cylinder	Ours, $\varepsilon = 2\%$	0.64	5.27	0.19	0.47	1.67

Table S5: Reference direction in 3-D (percent): mean order bound ν^*, mean NROE of our exact width-aware order, ST-MTM and TMTM (native orientation chosen in their favour), and mean attained displacement τ/L of our order with its width-aware lower bound, for the coordinate axes, the stored direction and the direction with the smallest bound among 400.

Sequence	Direction	ν^*	Ours	ST-MTM	TMTM	τ/L	bound
Tangaroa	x (1.00, 0.00, 0.00)	6.09	6.35	9.80	21.59	7.27	6.84
	y (0.00, 1.00, 0.00)	9.61	10.17	18.75	21.14	10.97	9.99
	z (0.00, 0.00, 1.00)	12.42	15.05	21.08	21.47	33.93	24.04
	stored (0.10, -0.99, -0.02)	9.24	9.67	15.80	19.50	10.37	9.54
	smallest bound (-0.61, -0.18, 0.77)	4.70	5.14	7.72	15.42	6.43	5.45
Cylinder	x (1.00, 0.00, 0.00)	7.72	8.01	17.77	26.62	8.47	8.35
	y (0.00, 1.00, 0.00)	8.20	8.47	17.62	17.95	10.58	9.53
	z (0.00, 0.00, 1.00)	33.61	33.88	39.92	39.55	38.58	34.75
	stored (1.00, 0.00, 0.00)	7.72	8.01	17.77	26.62	8.47	8.35
	smallest bound (-0.19, 0.98, 0.01)	5.26	5.86	13.89	16.14	6.91	6.18

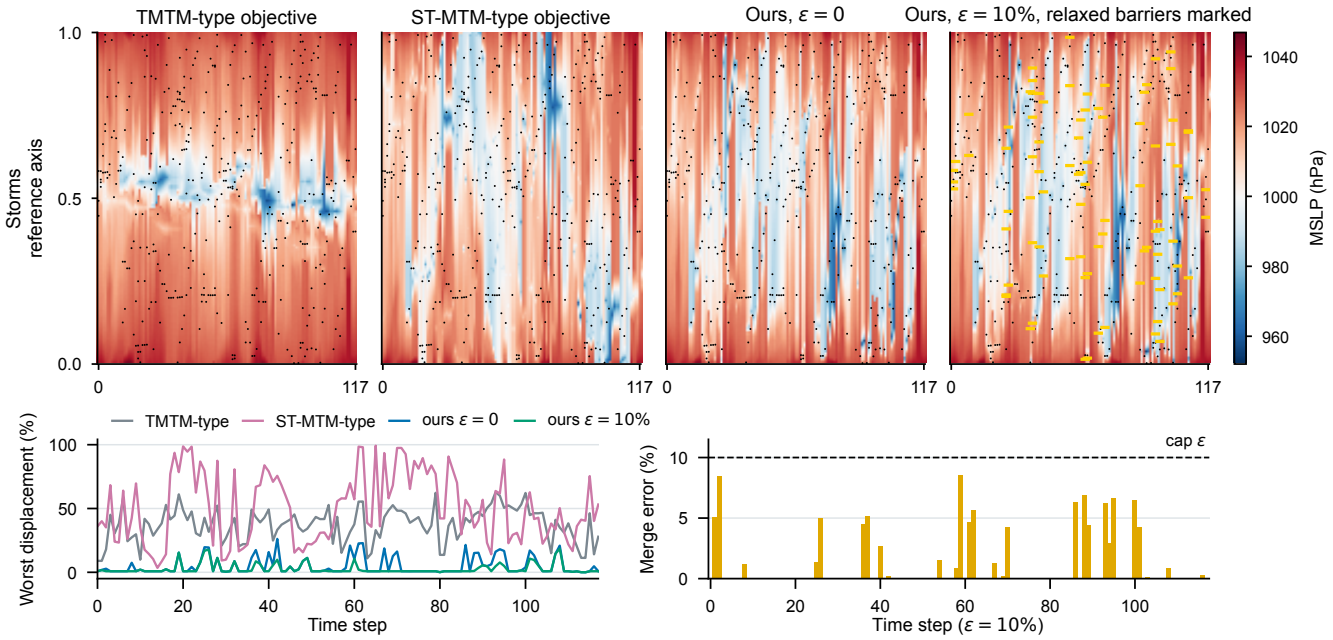


Figure S7: Storms, 118 half-day steps. Top: maps that differ only in the layout objective; dots mark references. On the $\varepsilon = 10\%$ map, yellow ticks mark barriers whose displayed merge level differs from the source by more than 0.1% of the scalar range. Bottom left: worst displacement per step. Bottom right: merge error spent per step at $\varepsilon = 10\%$, always below the cap.